Supplementary material for Reading Buys Evidence, Not a Decision: The Cost of Verification and the Limits of Retrieval

What this document is

The paper is at the conference page limit. The analyses added in response to review are reported in the paper itself; the full tables supporting them are here, together with the material Reviewer 1 asked for. Every number in this document is read from the retained result files at build time.

1 Reproduction

The analyses requested at review need per-claim predictions, which the original run did not retain. The study was therefore re-run from the same input file, whose SHA-256 digest was re-verified against the value the original manifest recorded, with per-claim predictions retained.

The re-run reproduced every published value in Sections 3 to 6 of the paper. Table S1 reconciles the accuracies recomputed from the per-claim records against the table the paper reports: they agree at every budget to four decimal places.

Table S1: Accuracies recomputed from the retained per-claim predictions against the values reported in the paper.

Sentences read	From per-claim records	Reported in the paper	Agrees
1	0.4001	0.4001	yes
2	0.4117	0.4117	yes
5	0.4080	0.4080	yes
10	0.4223	0.4223	yes
20	0.4229	0.4229	yes
50	0.4183	0.4183	yes
100	0.4281	0.4281	yes

1.1 What did not reproduce exactly

The language model arm did not reproduce bit for bit. It runs a 4-bit quantised model under greedy decoding through a GPU kernel library whose reduction order is not guaranteed to be stable across invocations, so a small number of completions differ between runs. Accuracy at 1 and 100 sentences reproduced exactly; at 5 and 20 it moved. Table S2 gives both runs. The paper reports the original run's values, which are the ones submitted and the ones its manifest records. The per-claim records used for the matched comparison in Section 7 come from the re-run, so that comparison is reported against the re-run's own accuracies rather than against the original ones. We record this rather than present the arm as exactly reproducible.

Table S2: The language model arm across the two runs. Accuracy is over all 198 subsampled claims; every claim was answered in both runs.

Sentences read	Original run	Re-run	Claims differing
1	0.4697	0.4697	0
5	0.4848	0.4697	3
20	0.4091	0.3990	2
100	0.3485	0.3485	0

2 Evidence recall under both definitions

Reviewer 3 asked whether recall counts a single retrieved annotator sentence or a complete annotator evidence set. The paper reported the first. Both are given here. A FEVER claim may be settled by any one of several annotator groups, so the complete-set measure asks whether at least one whole group lies inside the retrieved set, which is what the shared task's own score requires.

Table S3: Evidence recall under both definitions, over the 13,332 claims that can be settled. Claims labelled not settled have no annotator evidence by construction.

Sentences read	Settleable claims	At least one sentence	A complete set	Difference	Right page
1	13,332	0.5892	0.5390	0.0502	0.8372
2	13,332	0.7221	0.6669	0.0552	0.9122
5	13,332	0.8489	0.7895	0.0595	0.9609
10	13,332	0.9089	0.8521	0.0568	0.9791
20	13,332	0.9496	0.8977	0.0519	0.9876
50	13,332	0.9761	0.9302	0.0460	0.9939
100	13,332	0.9863	0.9453	0.0410	0.9965

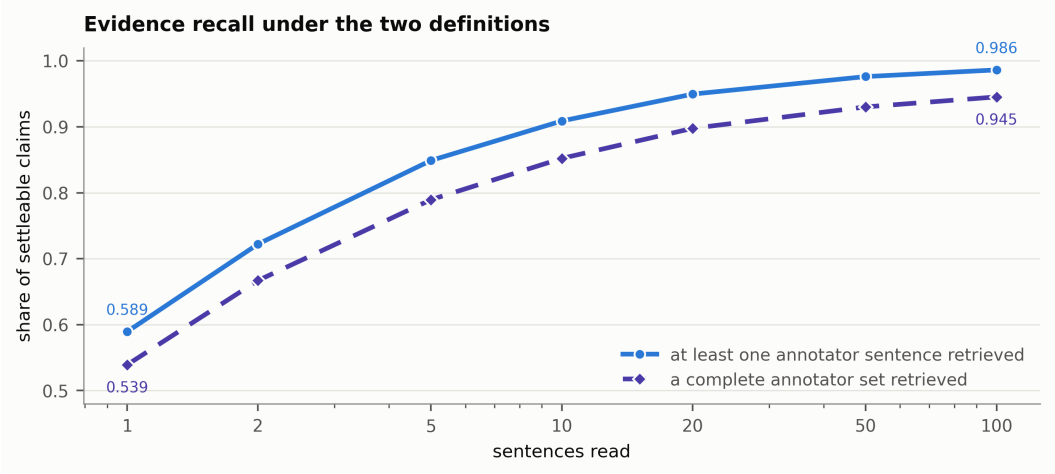


Figure S1: Evidence recall under the two definitions. The stricter curve runs below the other at every budget.

3 Paired comparisons between budgets

Every accuracy comparison the paper makes is between two conditions scored on the same 9,999 held-out claims, so the comparisons are paired. Intervals are percentile bootstrap intervals over 10,000 resamples of claims; the test is McNemar's exact test on the discordant pairs. Both are computed with seed 0.

Table S4: Budget against budget for the retrieval a system can actually run. The difference is the larger budget minus the smaller.

From	To	Accuracy from	Accuracy to	Difference	95% interval	Discordant	McNemar p
1	2	0.4001	0.4117	+0.0116	+0.0004 to +0.0231	3,356	0.0471
2	5	0.4117	0.4080	-0.0037	-0.0146 to +0.0071	3,187	0.5237
5	10	0.4080	0.4223	+0.0143	+0.0034 to +0.0253	3,099	0.0107
10	20	0.4223	0.4229	+0.0006	-0.0102 to +0.0115	3,006	0.9273
20	50	0.4229	0.4183	-0.0046	-0.0151 to +0.0061	2,976	0.4094
50	100	0.4183	0.4281	+0.0098	-0.0006 to +0.0201	2,758	0.0647
1	100	0.4001	0.4281	+0.0280	+0.0162 to +0.0396	3,640	<0.0001

Table S5: Budget against budget for the arm with the annotator's own sentences placed first. No consecutive difference is separable from zero except 50 to 100; nor is the whole range, which is why the paper declines to interpret the non-monotonicity. The interval over the full range runs from -0.0073 to +0.0163, so differences larger than about 0.016 in either direction are not compatible with these data. That bounds the dip rather than showing the two arms to be equal, which these data cannot show.

From	To	Accuracy from	Accuracy to	Difference	95% interval	Discordant	McNemar p
1	2	0.4254	0.4285	+0.0031	-0.0082 to +0.0141	3,191	0.5954
2	5	0.4285	0.4210	-0.0075	-0.0186 to +0.0033	3,131	0.1860
5	10	0.4210	0.4207	-0.0003	-0.0109 to +0.0106	2,957	0.9707
10	20	0.4207	0.4197	-0.0010	-0.0116 to +0.0095	2,924	0.8678
20	50	0.4197	0.4162	-0.0035	-0.0139 to +0.0072	2,947	0.5311
50	100	0.4162	0.4300	+0.0138	+0.0033 to +0.0241	2,820	0.0099
1	100	0.4254	0.4300	+0.0046	-0.0073 to +0.0163	3,692	0.4589

4 Annotator-first retrieval against feasible retrieval

Both arms are scored on the same claims at the same budget, so this comparison is paired claim by claim. The gain is confined to budgets below ten.

Table S6: Promoting every settleable claim's annotator sentences to the top of the retrieved list, against the retrieval a system can run, paired at each budget.

Sentences read	Feasible	Annotator-first	Difference	95% interval	Wrong to right	Right to wrong	McNemar p
1	0.4001	0.4254	+0.0253	+0.0162 to +0.0344	1,213	960	<0.0001
2	0.4117	0.4285	+0.0168	+0.0075 to +0.0261	1,219	1,051	0.0005
5	0.4080	0.4210	+0.0130	+0.0045 to +0.0215	1,018	888	0.0031
10	0.4223	0.4207	-0.0016	-0.0101 to +0.0070	916	932	0.7272
20	0.4229	0.4197	-0.0032	-0.0115 to +0.0050	846	878	0.4553
50	0.4183	0.4162	-0.0021	-0.0096 to +0.0054	720	741	0.6008
100	0.4281	0.4300	+0.0019	-0.0056 to +0.0094	719	700	0.6328

5 Where the accuracy gained by reading more comes from

Reviewer 3 asked why coverage rises while accuracy sometimes falls. The held-out claims are split by whether a complete annotator evidence set had been retrieved at the smaller budget, at the larger, at both or at neither. The paired accuracy change is computed within each group. Claims that cannot be settled have no annotator evidence, so they fall in the third group throughout.

The result is the one the paper reports: the gain is not concentrated where the evidence arrives. Over the full range the group whose evidence never arrives gains at least as much as the group whose evidence does, while the group that already had its evidence at one sentence does not move.

6 What survives resampling

Reviewer 3 asked for the non-monotonicity of the annotator-first arm to be explained. Intervals bound it but do not say what it is, so everything was held fixed except the seed, which controls both the train/test split and the classifier, then the study was repeated over 20 seeds. Features depend on the retrieval and the budget rather than the seed, so they were computed once and shared. Seed 0 is the split the paper reports and reproduces both arms exactly.

Three things follow. The second was not anticipated.

6.1 The dip the question points at is resampling variation

On the reported split the annotator-first minimum falls at 50 sentences. Over the 20 seeds it falls at 1 sentences on 9, 5 sentences on 3, 10 sentences on 1, 20 sentences on 3, 50 sentences on 3 and 100 sentences on 1 of them. The particular dip is a property of one split rather than of the data, which is why the paper declines to give it a mechanism.

Table S7: Paired accuracy change within each group. The final three rows span the whole range from one sentence to a hundred.

From	To	Group	Claims	Accuracy from	Accuracy to	Difference	95% interval	McNemar p
1	2	evidence arrived	860	0.3500	0.3895	+0.0395	-0.0023 to +0.0814	0.0717
1	2	already retrieved	3,625	0.4552	0.4394	-0.0157	-0.0342 to +0.0025	0.1008
1	2	never retrieved	5,514	0.3718	0.3970	+0.0252	+0.0096 to +0.0404	0.0013
2	5	evidence arrived	848	0.3325	0.3172	-0.0153	-0.0519 to +0.0224	0.4644
2	5	already retrieved	4,485	0.4299	0.4357	+0.0058	-0.0105 to +0.0221	0.4978
2	5	never retrieved	4,666	0.4087	0.3980	-0.0107	-0.0272 to +0.0058	0.2144
5	10	evidence arrived	450	0.3244	0.3822	+0.0578	+0.0089 to +0.1089	0.0292
5	10	already retrieved	5,333	0.4168	0.4213	+0.0045	-0.0101 to +0.0189	0.5633
5	10	never retrieved	4,216	0.4058	0.4279	+0.0221	+0.0050 to +0.0396	0.0133
10	20	evidence arrived	320	0.3000	0.3563	+0.0563	-0.0031 to +0.1156	0.0886
10	20	already retrieved	5,783	0.4183	0.4214	+0.0031	-0.0107 to +0.0169	0.6756
10	20	never retrieved	3,896	0.4384	0.4307	-0.0077	-0.0257 to +0.0098	0.4132
20	50	evidence arrived	195	0.2872	0.3231	+0.0359	-0.0359 to +0.1077	0.4101
20	50	already retrieved	6,103	0.4180	0.4073	-0.0107	-0.0243 to +0.0030	0.1257
20	50	never retrieved	3,701	0.4383	0.4415	+0.0032	-0.0149 to +0.0213	0.7484
50	100	evidence arrived	99	0.4444	0.4747	+0.0303	-0.0808 to +0.1414	0.7201
50	100	already retrieved	6,298	0.4047	0.4128	+0.0081	-0.0041 to +0.0206	0.2111
50	100	never retrieved	3,602	0.4414	0.4536	+0.0122	-0.0061 to +0.0300	0.2004
1	100	evidence arrived	2,772	0.3243	0.3662	+0.0418	+0.0198 to +0.0642	0.0002
1	100	already retrieved	3,625	0.4552	0.4502	-0.0050	-0.0240 to +0.0141	0.6273
1	100	never retrieved	3,602	0.4031	0.4536	+0.0505	+0.0303 to +0.0711	<0.0001

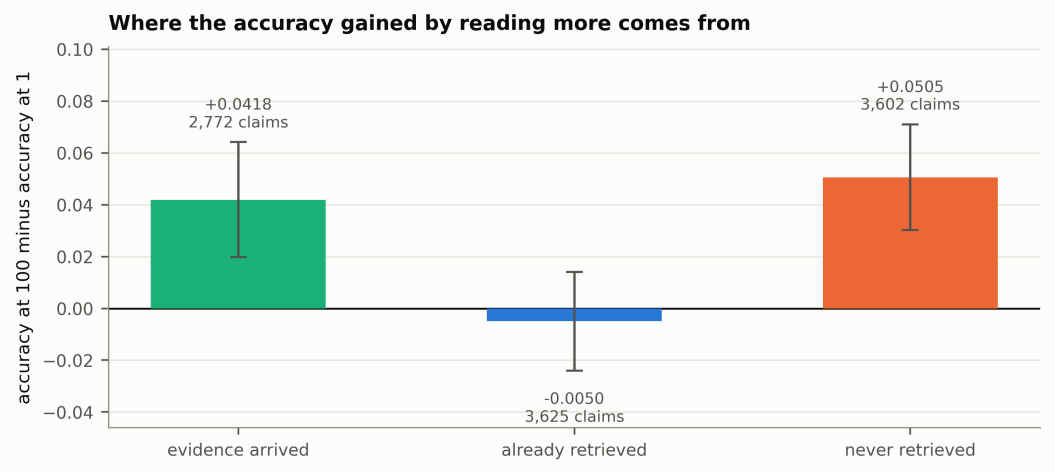


Figure S2: Accuracy change from one sentence to a hundred within each group, with paired bootstrap intervals.

6.2 A different non-monotonicity is systematic

The annotator-first arm rises from one sentence to two on every seed and falls from two to five on all but one. That is not noise. The arm a system can actually run does not do this: its two-to-five step rises on most seeds. The drop therefore belongs to the intervention rather than to the budget. A reading consistent with the contrast, which this study does not test, is that once the annotator sentences are placed first every further sentence is a non-gold one, so the summary statistics the verifier reads are diluted by material that cannot settle the claim.

Table S8: Budget against budget within each seed, averaged over 20 seeds. Improving counts the seeds on which the larger budget scored higher.

Arm	From	To	Difference	SD	Improving
annotator-first	1	2	+0.0102	0.0044	20 of 20
annotator-first	2	5	-0.0068	0.0049	1 of 20
annotator-first	5	10	-0.0008	0.0056	7 of 20
annotator-first	10	20	-0.0014	0.0055	6 of 20
annotator-first	20	50	+0.0011	0.0061	13 of 20
annotator-first	50	100	+0.0045	0.0062	14 of 20
annotator-first	1	100	+0.0067	0.0056	16 of 20
feasible	1	2	+0.0141	0.0045	20 of 20
feasible	2	5	+0.0041	0.0063	15 of 20
feasible	5	10	+0.0010	0.0065	13 of 20
feasible	10	20	+0.0001	0.0051	9 of 20
feasible	20	50	+0.0035	0.0066	13 of 20
feasible	50	100	+0.0049	0.0048	16 of 20
feasible	1	100	+0.0276	0.0049	20 of 20

6.3 The reported split is a favourable one for this arm

The gain from placing the annotator sentences first is larger on the reported split than on most others at the small budgets, being smaller at ten and twenty. The direction of the result is unchanged: the gain is largest when little is read and decays towards zero. What resampling does not support is the sharper reading that it is confined below five sentences and absent thereafter, because across seeds it declines smoothly and remains slightly positive at ten and twenty. The paper states the reported-split values, which were produced by the seed fixed before the run; this table is the qualification.

Table S9: The annotator-first gain on the reported split against its distribution over 20 seeds.

Sentences read	Reported split	Mean of 20 seeds	SD	Seeds above the reported value
1	+0.0253	+0.0197	0.0040	3 of 20
2	+0.0168	+0.0159	0.0049	6 of 20
5	+0.0130	+0.0049	0.0047	0 of 20
10	-0.0016	+0.0031	0.0036	17 of 20
20	-0.0032	+0.0016	0.0038	17 of 20
50	-0.0021	-0.0008	0.0052	11 of 20
100	+0.0019	-0.0011	0.0036	4 of 20

The central result of the paper is not affected. In the arm a system can run, accuracy from one sentence to a hundred rises by +0.0276 averaged over the 20 seeds, improving on every one, against the +0.0280 the paper reports on its own split.

7 The language model comparison, matched

Reviewer 3 observed that the language model was scored on 198 claims while the feature verifier was scored on 9,999. The subsample is drawn deterministically from the same held-out half, stratified by label, so it was reproduced exactly and the feature verifier rescored on precisely those 198 claims. Their identifiers are retained with the artifacts.

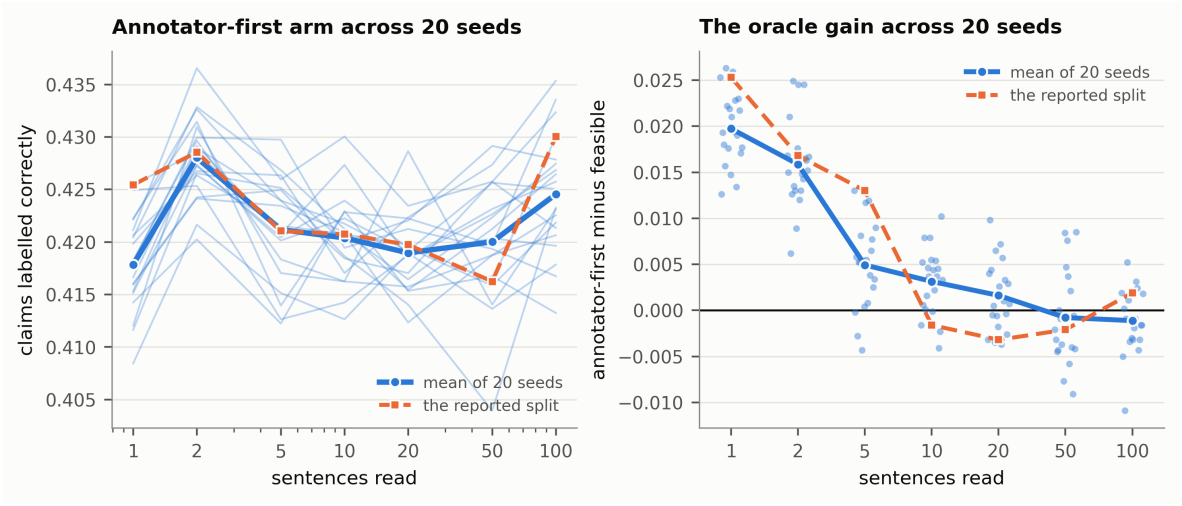


Figure S3: Left: the annotator-first arm on each of the 20 seeds, with their mean and the reported split. Right: the gain over feasible retrieval at each budget, one point per seed.

Table S10: The two verifiers on exactly the same 198 claims, with the full held-out half shown for reference. The language model column is the re-run, because the matched test needs its per-claim records.

Sentences read	Language model (198)	Feature verifier (198)	Its 95% interval	Feature verifier (9,999)
1	0.4697	0.3788	0.3131 to 0.4495	0.4001
5	0.4697	0.3990	0.3333 to 0.4697	0.4080
20	0.3990	0.4091	0.3434 to 0.4798	0.4229
100	0.3485	0.3737	0.3081 to 0.4444	0.4281

Table S11: Paired test between the language model and the feature verifier on the same 198 claims. The difference is the language model minus the feature verifier.

Sentences read	Difference	95% interval	LM right only	Features right only	McNemar p
1	+0.0909	+0.0000 to +0.1818	52	34	0.0662
5	+0.0707	-0.0253 to +0.1616	51	37	0.1654
20	-0.0101	-0.1010 to +0.0808	44	46	0.9161
100	-0.0253	-0.1212 to +0.0707	45	50	0.6817

8 The prompt, verbatim

Reviewer 1 noted that the exact prompts were not retained. The template is reproduced here exactly as it appears in the source. The placeholders are the claim and the retrieved sentences at the budget being evaluated; each sentence is truncated to 320 characters and prefixed with a hyphen. The assembled user message is then wrapped by the model's own chat template with thinking disabled and generation primed.

The per-claim prompts themselves are also supplied, all 792 of them, one per claim per budget. They were not retained by the original run and are not recovered from a log: the prompt is a deterministic function of the claim text and the retrieved sentences, so each is RECONSTRUCTED from the retained ranking and sentence store by the same function the run calls. The digests of both inputs are recorded with them. Reconstructing the prompt does not reconstruct the completion; the completions supplied are the re-run's.

```
Decide whether the evidence supports the claim, refutes it or is insufficient.

Claim: {claim}

Evidence:
- {sentence 1}
- {sentence 2}
  ...up to the budget; with no retrieved sentence the list is
  the single line "- (no evidence retrieved)"

Answer with exactly one of these three labels and nothing else: SUPPORTS, REFUTES, NOT ENOUGH
    ↪ INFO.
```

The decoding settings are unchanged from the original run: model `mlx-community/SmolLM3-3B-4bit`, greedy decoding, at most 8 new tokens, budgets 1, 5, 20, 100, seed 0. A completion is read for a label by the parser in the source; a completion naming no label counts as unanswered. Every one of the 198 claims was answered at every budget in both runs.

9 Artifacts

The bundle accompanying this submission is anonymous. It contains the study source, the run manifests with the input digest and the pool sampling record, the retrieval index of the top 100 sentences for every claim, per-claim predictions for both arms at every budget, the per-claim evidence-retrieval record used in Table S7, the language model per-claim outputs from the re-run, the subsample identifiers, the reconstructed per-claim prompts with their manifest and every table in this document as CSV.

The retrieval pool is rebuilt from the published corpus by the included script rather than shipped, because it is derived from a 1.7 GB third-party snapshot. The script records the digest of what it downloads.

10 Figures held here rather than in the paper

The paper is at the page limit, so two figures were moved here. Neither carries data the paper lacks: Fig. S4 plots the recall columns of the paper's own Table 1 together with the pool composition its text gives in numbers; Fig. S5 plots the two accuracy series of the paper's own Table 2.

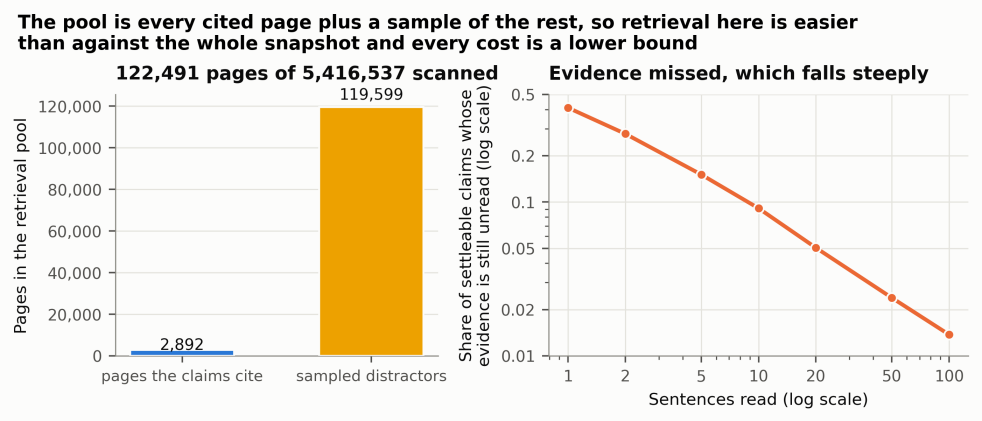


Figure S4: The retrieval pool and the share of settleable claims whose evidence is still unread at each budget.

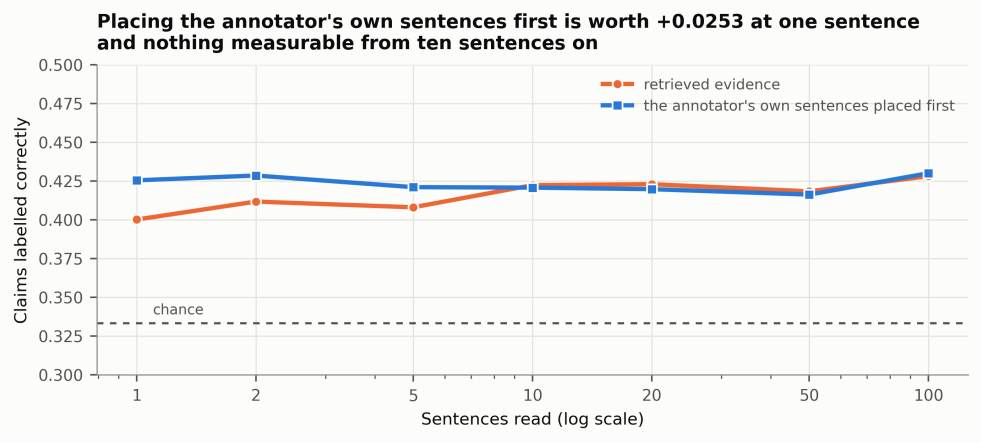


Figure S5: Verification with retrieved evidence against the annotator's own sentences placed first.